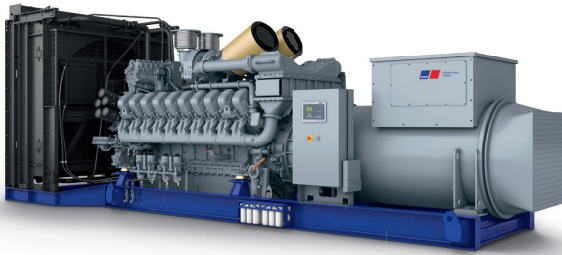




Diesel Generator Set

mtu 20V4000 DS3600

400 V - 11 kV/50 Hz/data center continuous power/
fuel consumption optimized/20V4000G44F/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

Product highlights

Benefits

- Low fuel consumption
- Optimized system integration ability
- High reliability
- High availability of power
- Long maintenance intervals

Support

- Global product support offered

Standards

- Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
- Generator set complies to ISO 8528
- Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
- NFPA 110

Power rating

- System ratings: 3370 kVA - 3390 kVA
- Accepts rated load in one step per NFPA 110*
- Generator set complies to G3 according to ISO 8528-5
- Generator set exceeds load steps according to ISO 8528-5*

Performance assurance certification (PAC)

- Engine-generator set tested to ISO 8528-5 for transient response
- 100% load factor
- Verified product design, quality and performance integrity
- All engine systems are prototype and factory tested

Complete range of accessories available

- Control panel
- Power panel
- Fuel system
- Fuel connections with shut-off valve mounted to base frame
- Starting/charging system
- Exhaust system
- Electrical driven radiators
- Mechanical driven radiators
- Medium and oversized voltage alternators
- Low voltage alternator

Emissions

- Fuel consumption optimized

Certifications

- CE certification option
- Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary



A Rolls-Royce
solution

Application data ¹⁾

Engine			Liquid capacity (lubrication)	
Manufacturer		<i>mtu</i>	Total oil system capacity: l	390
Model	20V4000G44F		Engine jacket water capacity: l	260
Type	4-cycle		Intercooler coolant capacity: l	50
Arrangement	20V		Combustion air requirements	
Displacement: l	95.4		Combustion air volume: m³/s	4.0
Bore: mm	170		Max. air intake restriction: mbar	30
Stroke: mm	210		Cooling/radiator system	
Compression ratio	16.4		Coolant flow rate (HT circuit): m³/hr	80
Rated speed: rpm	1500		Coolant flow rate (LT circuit): m³/hr	44
Engine governor	ECU 9		Heat rejection to coolant: kW (100/110%)	945/1090
Max power: kWm	2807		Heat radiated to charge air cooling: kW (100/110%)	700/795
Air cleaner	dry		Heat radiated to ambient: kW	105
Fuel system			Fan power for electr. radiator (40°C): kW	105
Maximum fuel lift: m	5		Exhaust system	
Total fuel flow: l/min	27		Exhaust gas temp. (after engine, max.): °C	550
Fuel consumption ²⁾			Exhaust gas temp. (before turbocharger): °C	605
At 100% of power rating:	653	193	Exhaust gas volume: m³/s	9.6
At 75% of power rating:	485	191	Maximum allowable back pressure: mbar	50
At 50% of power rating:	349	206	Minimum allowable back pressure: mbar	-

Standard and optional features

System ratings (kW/kVA)

Generator model	Voltage	fuel consumption optimized 40°C/1000m		
		without radiator		
		kWel	kVA*	AMPS
Leroy Somer LSA54.2 ZL17 (LV Leroy Somer standard)	400 V	2696	3370	4864
Leroy Somer LSA54.2 XL11 (Med. volt. Leroy Somer)	11 kV	2704	3380	177
Marathon 1040FDH7103 (Medium volt. marathon)	11 kV	2712	3390	178
Leroy Somer LSA54.2 ZL12 (MV Leroy Somer oversized)	11 kV	2704	3380	177
Marathon 1040FDH7105 (MV marathon oversized)	11 kV	2712	3390	178

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).
2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power and are approximate values.